

GOULBURN, Australia

WMO: 957160

Lat: 34.8086S

Lon: 149.7311E

Elev: 641

StdP: 93.86

Time Zone: 10.00 (AUS)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	-4.6	-3.1	-7.3	2.2	4.8	-5.4	2.6	3.9	14.6	7.8	13.5	8.3	0.4	0	0.596

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	14.1	33.8	17.6	31.5	17.1	29.2	16.6	20.2	26.9	19.4	25.7	18.6	24.7	6.8	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.5	14.5	21.6	17.6	13.6	21.0	16.7	12.9	20.3	60.9	26.9	58.0	25.8	55.3	24.6	30.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.1	11.6	10.2	-7.4	37.0	1.4	2.2	-8.5	38.6	-9.3	39.9	-10.1	41.1	-11.1	42.7		
(5)			-7.9	21.9	1.5	2.0	-8.9	23.3	-9.8	24.5	-10.6	25.6	-11.7	27.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	12.7	20.0	19.3	16.6	12.8	9.0	6.7	5.8	6.8	9.5	12.4	15.4	18.2
	DBStd	5.80	3.56	3.26	3.06	2.99	2.72	2.41	2.27	2.54	2.75	3.23	3.75	3.51
	HDD10.0	456	0	0	1	9	52	102	131	105	41	13	3	0
	HDD18.3	2269	23	24	69	167	289	349	388	358	264	187	102	47
	CDD10.0	1432	311	261	206	93	21	3	1	5	27	86	166	253
	CDD18.3	202	76	52	17	1	0	0	0	0	1	14	41	
	CDH23.3	2817	983	557	236	34	0	0	0	0	4	74	307	622
	CDH26.7	1119	467	222	67	6	0	0	0	0	1	11	108	238
Wind	WSAvg	4.2	4.4	4.1	3.9	3.3	3.7	3.8	4.4	4.7	4.8	4.5	4.5	4.5
Precipitation	PrecAvg	628	61	59	55	50	46	46	45	57	50	58	66	56
	PrecMax	911	181	167	181	208	125	185	97	215	98	148	117	131
	PrecMin	362	3	2	2	0	3	9	4	5	4	5	5	1
	PrecStd	148	41	46	40	53	35	38	25	49	24	33	32	38
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.1	35.0	31.7	27.6	20.7	16.5	15.8	19.6	23.3	28.3	33.4	35.0
		MCWB	18.1	18.4	16.2	15.2	12.2	10.7	10.3	11.2	12.5	14.5	16.0	16.7
	2%	DB	34.5	32.2	28.4	23.7	18.6	14.2	13.7	16.5	20.8	25.4	29.6	31.9
		MCWB	18.0	17.6	16.1	14.0	12.3	10.1	8.9	9.6	12.0	14.0	15.9	16.6
	5%	DB	31.9	29.3	26.0	21.7	16.8	12.8	12.1	14.5	18.7	23.0	26.9	29.4
		MCWB	17.5	17.4	15.8	13.5	11.4	9.5	8.2	8.9	11.3	13.4	15.5	16.0
	10%	DB	29.0	26.8	23.5	19.6	15.0	11.6	10.7	12.7	16.6	20.6	24.2	27.0
		MCWB	17.4	17.3	15.6	12.9	10.8	8.9	7.5	8.3	10.6	12.8	14.5	15.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.4	21.5	20.0	17.0	15.1	12.6	11.9	12.6	14.6	17.0	19.0	20.0
		MCDB	29.5	28.7	23.8	22.6	17.0	14.6	13.9	16.9	19.5	23.5	25.8	26.3
	2%	WB	20.3	20.2	18.6	15.7	13.6	11.2	9.9	10.8	13.3	15.6	17.6	18.8
		MCDB	28.2	26.0	23.4	20.7	16.4	12.9	12.1	14.8	18.3	21.6	24.9	25.2
	5%	WB	19.5	19.3	17.6	14.7	12.4	10.2	8.9	9.7	12.2	14.6	16.6	17.8
		MCDB	26.8	24.6	22.3	18.8	15.5	12.1	11.1	13.3	16.9	20.4	23.5	24.8
	10%	WB	18.6	18.4	16.7	13.8	11.3	9.2	8.0	8.6	11.2	13.6	15.7	17.0
		MCDB	25.4	23.7	21.1	17.9	14.3	11.0	10.2	12.1	15.4	19.1	21.7	23.9
Mean Daily Temperature Range	5% DB	MDBR	14.1	12.5	12.3	13.2	12.3	10.1	10.3	11.7	12.7	13.7	13.7	14.2
		MCDBR	20.6	17.9	17.3	16.6	15.5	12.6	13.1	15.2	17.3	18.7	19.6	19.2
		MCWBR	6.3	5.8	6.7	7.9	9.3	8.7	8.8	9.1	9.4	8.7	7.7	6.3
	5% WB	MCDBR	15.1	13.4	12.5	12.9	11.9	9.7	10.1	12.8	13.9	14.9	15.4	14.6
		MCWBR	5.7	5.3	5.8	6.8	7.6	7.1	7.3	8.2	8.1	7.9	7.0	5.8
Clear-Sky Solar Irradiance	taub	0.364	0.352	0.320	0.298	0.284	0.274	0.268	0.277	0.305	0.323	0.346	0.340	
	taud	2.438	2.503	2.575	2.614	2.603	2.641	2.657	2.617	2.520	2.489	2.459	2.502	
	Ebn at Noon	971	963	959	927	888	873	901	938	959	981	984	1000	
	Edh at Noon	121	109	94	80	71	64	66	78	98	109	118	114	
All-Sky Solar Radiation	RadAvg	6.93	5.91	5.00	3.93	3.00	2.29	2.63	3.52	4.85	5.90	6.41	7.07	
	RadStd	0.53	0.53	0.44	0.34	0.16	0.18	0.19	0.23	0.36	0.51	0.55	0.51	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.44	N/A	N/A	N/A	N/A	N/A	N/A	-96	+154	+63
(47) Regional (35 neighbors)		+0.43	N/A	N/A	N/A	N/A	N/A	-1	-83	+174	+82

Nomenclature: See separate page